

IEEE 754 Tutorial - Solutions

Question 1: Decimal 25.25 → IEEE 754 (8 hex characters)

Answer: **41ca0000**

Explanation: $25.25 = 1.100101 \times 2^4$; exponent = 131 (10000011); mantissa = 100101000... → Hex = 41ca0000.

Question 2: Signed decimal -19.75 → IEEE 754 (8 hex characters)

Answer: **c19e0000**

Explanation: $-19.75 = 1.001111 \times 2^4$; exponent = 131; sign = 1; mantissa = 001111000... → Hex = c19e0000.

Question 3: IEEE 754 representation C2B00000 → Decimal

Answer: **-88.0**

Explanation: Interpreting c2b00000 as IEEE 754 single precision gives -88.0.

Question 4: IEEE 754 representation C2BC0000 → Decimal

Answer: **-94.0**

Explanation: Interpreting c2bc0000 as IEEE 754 single precision gives -94.0.

Question 5: The denary value of IEEE 754 format 43F8000 is +63.75. True or False?

Answer: **False**

Explanation: 43F8000 is not a valid full 32-bit hex (missing one digit). If interpreted as 043f8000, it represents a very small value ($\sim 2.25 \times 10^{-3}$), not 63.75. The correct hex for +63.75 is 427f0000.